

Name:.....



HSC Mathematics

Assessment Task 1 - 2013

Time Allowed - 60 minutes + 5minutes reading

Instructions: Calculators may be used in any parts of the task. For 1 Mark Questions, the correct answer is sufficient to receive full marks. For Questions worth more than 1 Mark, necessary working MUST be shown to receive full marks.

Multiple Choice	
Questions 1 - 5	/5
Question 6	/12
Question 7	/11
Question 8	/12
Question 9	/11
Total	/51

Circle the correct answer - Questions 1 - 5 are worth 1 mark each

1. A parabola has a focus of $(0, -a)$ and a directrix at $y = a$. Its equation is:

A $y^2 = 4ax$

B $y^2 = -4ax$

C $x^2 = 4ay$

D $x^2 = -4ay$

2. The derivative of $\frac{1}{2x^2}$ is

A $-4x^{-3}$

B $\frac{1}{4x}$

C $\frac{-1}{x^3}$

D $\frac{-1}{2x^3}$

3. A parabola is the locus of points equidistant from:

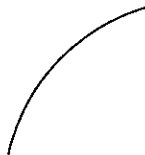
A two points

B two intersecting lines

C a point and a line

D two parallel lines

4. For the curve,



A $f'(x) > 0$ and $f''(x) < 0$

B $f'(x) > 0$ and $f''(x) > 0$

C $f'(x) < 0$ and $f''(x) < 0$

D $f'(x) < 0$ and $f''(x) > 0$

5. The gradient of the normal to $y = 2x^3 - 3x - 5$ at $(-1, -4)$ is

A -2

B 3

C $\frac{1}{2}$

D $-\frac{1}{3}$

Question 6 12 Marks (Begin a new sheet of paper)

Marks

a) Differentiate i) $y = 4x^3 - 7x - 2$

1

ii) $y = x^2\sqrt{x}$

2

b) Find the derivative in simplest factored form of

i) $y = (7 - 5x)^4$

2

ii) $y = \frac{(3x - 4)^4}{(2x + 3)^2}$

4

c) Differentiate from first principles $f(x) = 5 - 2x - 3x^2$

3

Question 7 12 Marks (Begin a new sheet of paper)

a) For the curve $y = 4x^3 - x^4$ find

i) The x intercepts **1**

ii) The coordinates and nature of the stationary points **4**

iii) Any points of inflexion **2**

iv) Sketch the curve **2**

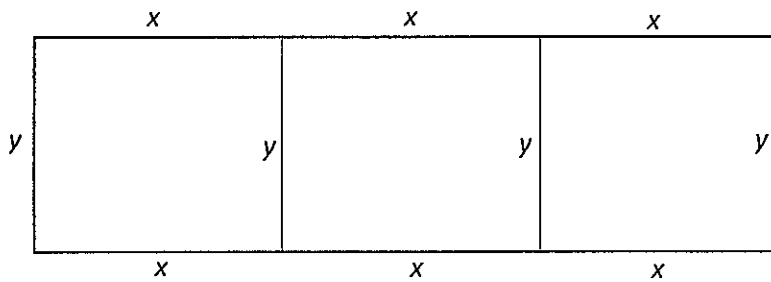
b) Find the equation of the tangent to the curve $y = 2\sqrt{2x+5}$ at the point where $x = 2$ **2**

Question 8 12 Marks (Begin a new sheet of paper)

- a) Find the equation of the locus of a point which is always 5 units from the point $(-1, 2)$ 1
- b) i) Sketch the straight lines $3x - 4y = 12$ and $5x + 12y = 12$ on the same number plane. Clearly label each line and your number plane should be at least $\frac{1}{3}$ page in size. 2
- ii) Find the equation of the locus of a point which is equidistant from these two lines. 4
- iii) Sketch and clearly label the locus on the same number plane as in i).. 1
- c) For the parabola $(x - 1)^2 = 4(y + 2)$ find 1
- i) The coordinates of the vertex 1
- ii) The coordinates of the focus 1
- iii) The equation of the dirextrix 1
- iv) Sketch the parabola 1

Question 9 (11 Marks) (Begin a new sheet of paper)

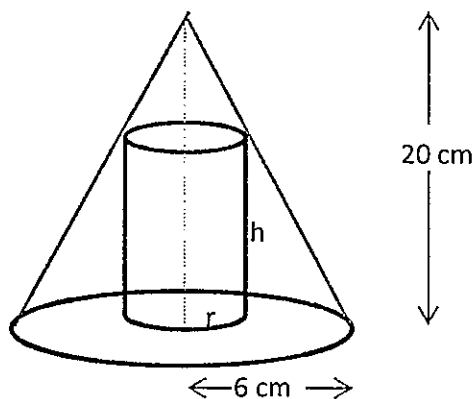
- a) A 20 cm piece of wire is cut to form the shape below



- i) Show that $y = 5 - \frac{3}{2}x$. 1
- ii) Show that the total enclosed area is given by $A = 15x - \frac{9}{2}x^2$ 1
- iii) Find the value of x such that the enclosed area is a maximum. 3

- b) A cylinder of radius r cm and height h cm is inscribed in a cone with base radius 6 cm and height 20 cm.

- i) Use similar triangles to show that $h = \frac{60-10r}{3}$ 2
- ii) Show that the volume of the cylinder is given by $V = \frac{10\pi r^2(6-r)}{3}$ 1
- iii) Hence find the values of r and h for the cylinder which has maximum volume. 3

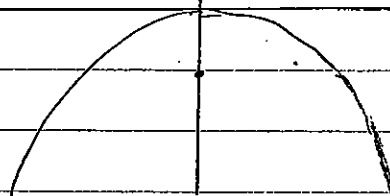


END OF TEST

Solutions to 2013 HSC Mathematics Task 1

1. $x^2 = -4ay$

(D)



2.

$$y = \frac{1}{2}x^{-2}$$

$$y' = \frac{1}{2} \cdot -2x^{-3} = -\frac{1}{x^3} \quad (C)$$

3.

(C)

4.

Increasing $\therefore f'(x) > 0$

Decrease Down $\therefore f''(x) < 0$

(A)

5.

$$y = 2x^3 - 3x - 5$$

$$-2 + 3 - 5 = -4$$

$$y' = 6x^2 - 3$$

$$m_{\text{tangent}} = 6 - 3 = 3$$

$$m_{\text{normal}} = -\frac{1}{3}$$

(D)

Question 6.

a) i) $y' = 12x^2 - 7$

ii) $y = x^{\frac{5}{2}}$

$$y' = \frac{5}{2} x^{\frac{3}{2}} = \frac{5}{2} x \sqrt{x}$$

b) i) $y' = 4(7-5x)^3 \cdot -5$
 $y' = -20(7-5x)^3$

ii) $y' = \frac{(2x+3)^2 \cdot 4(3x-4)^3 \cdot 3 - (3x-4)^4 \cdot 2(2x+3) \cdot 2}{(2x+3)^4}$

$$y' = \frac{4(2x+3)(3x-4)^3 \{3(2x+3) - (3x-4)\}}{(2x+3)^4}$$

$$y' = \frac{4(3x-4)^3(3x+13)}{(2x+3)^4}$$

c) $f(x) = 5 - 2x - 3x^2$

$$f(x+h) = 5 - 2(x+h) - 3(x+h)^2$$

$$= 5 - 2x - 2h - 3x^2 - 6xh - 3h^2$$

$$\frac{f(x+h) - f(x)}{h} = \frac{-2h - 6xh - 3h^2}{h}$$

$$\frac{f(x+h) - f(x)}{h} = -2 - 6x - 3h$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = -2 - 6x$$

7b) $y = 2(2x+5)^{\frac{1}{2}}$

$$y' = 2 \cdot \frac{1}{2} (2x+5)^{-\frac{1}{2}} \cdot 2 = \frac{2}{\sqrt{2x+5}}$$

$$x=2, \quad y=6, \quad m = \frac{2}{3}$$

$$\therefore y - 6 = \frac{2}{3}(x - 2)$$

$$3y - 18 = 2x - 4$$

$$2x - 3y + 14 = 0$$

Q7

$$a) \quad y = 4x^3 - x^4$$

$$y = x^3(4-x)$$

$$i) \quad y=0 \Rightarrow x=0 \text{ or } 4$$

$$ii) \quad y' = 12x^2 - 4x^3 \\ = 4x^2(3-x)$$

$$y' = 0 \Rightarrow \left. \begin{matrix} x=0 \\ y=0 \end{matrix} \right\} \text{ or } \left. \begin{matrix} x=3 \\ y=27 \end{matrix} \right\}$$

Nature of (0, 0)

x	0 ⁻	0	0 ⁺
y'	+	0	+

horizontal point of inflexion

Nature of (3, 27)

x	3 ⁻	3	3 ⁺
y'	+	0	-

Maximum turning point

ii) For pts of inflexion $y''=0$ + changes sign

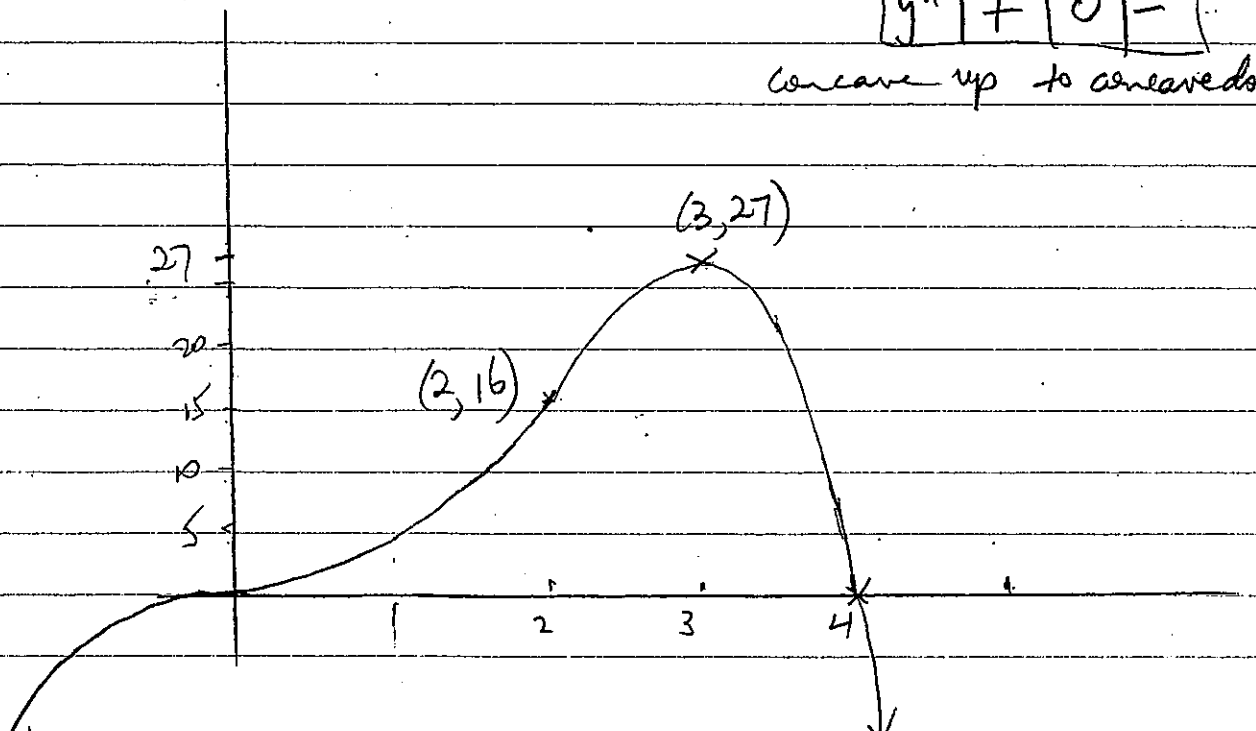
$$y'' = 24x - 12x^2 \\ = 12x(2-x)$$

$$\left. \begin{matrix} x=0 \\ y=0 \end{matrix} \right\} \text{ or } \left. \begin{matrix} x=2 \\ y=16 \end{matrix} \right\} \text{ are possible pts of inflexion}$$

already examined

x	2 ⁻	2	2 ⁺
y''	+	0	-

concave up to concave down

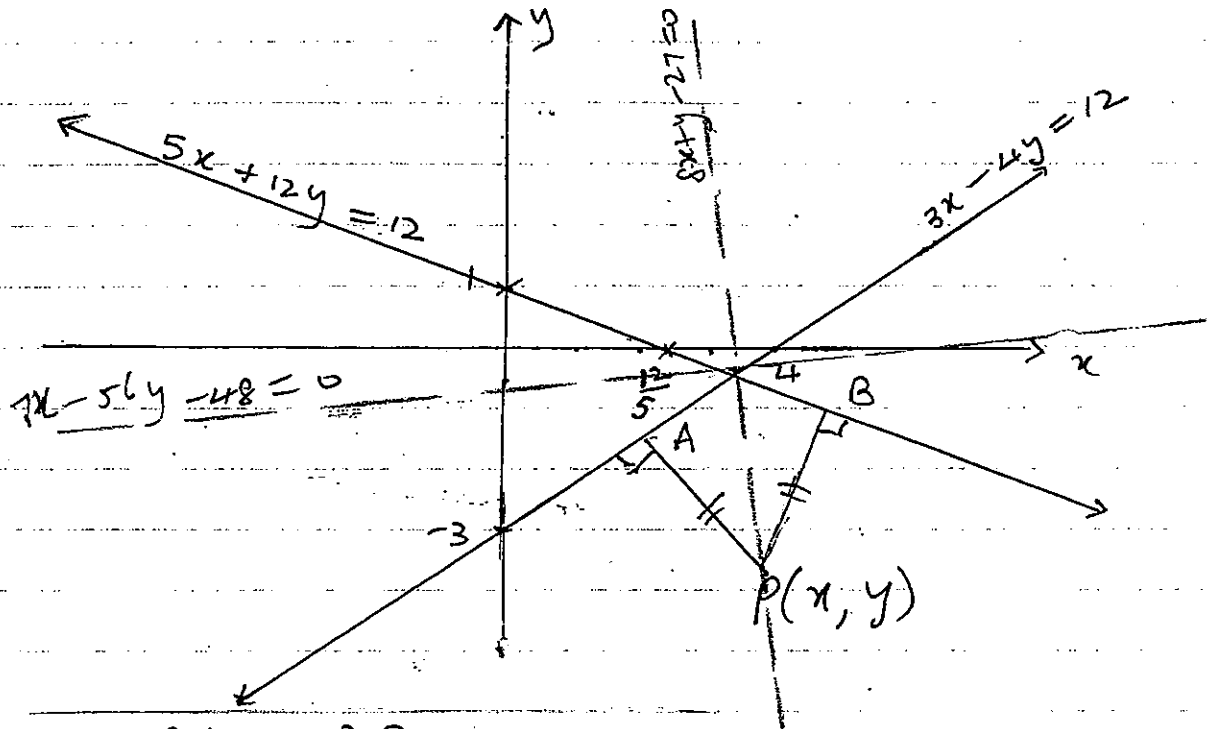


Question 8

a)

$$(x+1)^2 + (y-2)^2 = 25$$

b)



$$PA = PB$$

$$\left| \frac{3x - 4y - 12}{5} \right| = \left| \frac{5x + 12y - 12}{13} \right|$$

$$39x - 52y - 156 = 25x + 60y - 60 \quad \text{or} \quad 39x - 52y - 156 = -25x - 60y + 60$$

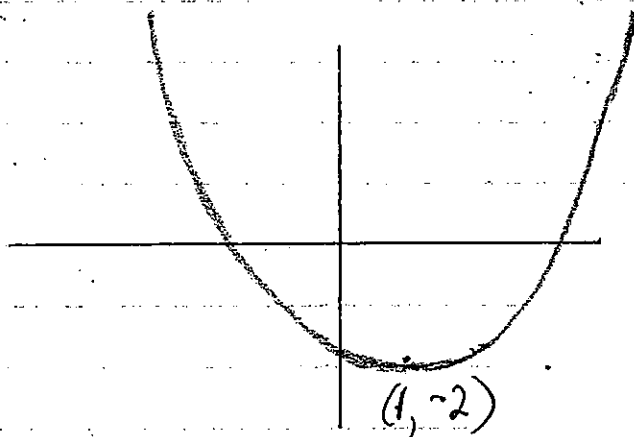
$$14x - 112y - 96 = 0$$

$$\checkmark \quad 64x + 8y - 216 = 0$$

$$7x - 56y - 48 = 0$$

$$\checkmark \quad 8x + y - 27 = 0$$

c)



i) vertex $(1, -2)$

ii) focus $(1, -1)$

iii) directrix
 $y = -3$

$$a = 1$$

$$\begin{aligned}
 \text{9 a) i)} \quad 6x + 4y &= 20 \\
 3x + 2y &= 10 \\
 2y &= 10 - 3x \\
 y &= 5 - \frac{3x}{2}
 \end{aligned}$$

$$\text{ii)} \quad A = 3x \times y$$

$$A = 3x \left(5 - \frac{3x}{2} \right)$$

$$A = 15x - \frac{9x^2}{2}$$

$$\text{iii)} \quad \frac{dA}{dx} = 15 - \frac{18x}{2}$$

For maximum area, $\frac{dA}{dx} = 0$

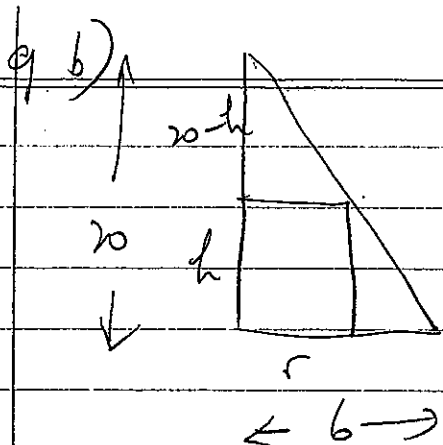
$$15 - 9x = 0$$

$$x = \frac{15}{9} = \frac{5}{3}$$

Show that A is in fact maximum

$$\frac{d^2A}{dx^2} = -9$$

which is negative $\therefore A$ is maximum



$$i) \quad \frac{20-h}{20} = \frac{r}{6}$$

$$20r = 6(20-h)$$

$$10r = 3(20-h)$$

$$10r = 60 - 3h$$

$$3h = 60 - 10r$$

$$h = \frac{60 - 10r}{3}$$

$$h = \frac{10}{3}(6-r)$$

$$ii) \quad V = \pi r^2 h$$

$$V = \pi r^2 \cdot \frac{10}{3}(6-r)$$

$$= \frac{10\pi r^2}{3}(6-r)$$

$$iii) \quad V = \frac{10\pi}{3}(6r^2 - r^3)$$

$$\frac{dV}{dr} = 0 \text{ for maximum volume}$$

$$\frac{10\pi}{3}(12r - 3r^2) = 0$$

$$3r(4-r) = 0$$

$$r = 0 \text{ or } 4$$

Show that V is in fact a maximum when $r = 4$

$$\frac{d^2V}{dr^2} = \frac{10\pi}{3}(12 - 6r)$$

$$= \frac{10\pi}{3}(12 - 24) \text{ which is -ve}$$

$\therefore$ because down $\therefore V$ is maximum

$$\text{When } r = 4, \quad h = \frac{10}{3}(6-4)$$

$$r = 4, \quad h = \frac{20}{3}$$